Laboratory Practice: Rotating Spring Dynamics

1 Objective

The objective of this experiment is to study the dynamics of a rotating spring. Specifically, to analyze its radial extension under rotation, determining the effective elastic-inertial parameters, and to measure the "collapse time" of the spring upon release, interpreting it in terms of wave propagation.

2 Material

1. Rotating motor system (Rotor radius r_0)
2. Helical Spring (Mass M, Constant K)
3. High-speed camera system
4. Computer with video tracking software
5. Dataset files (Dataset A and Dataset B)

3 Theoretical Background

A spring of total mass M and effective spring constant K is attached at its inner end to a rotor of radius r_0 and rotated at constant angular speed ω in a horizontal plane. The spring stretches radially and reaches a steady configuration.

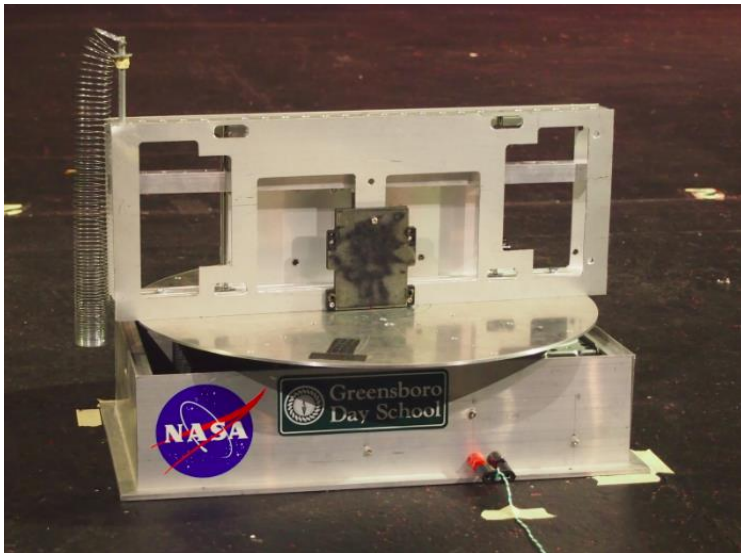


Figure 1: The experimental apparatus used to rotate the spring.

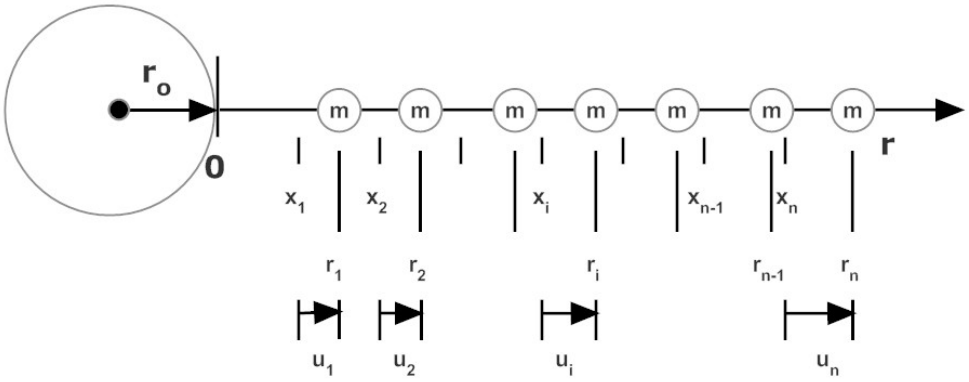
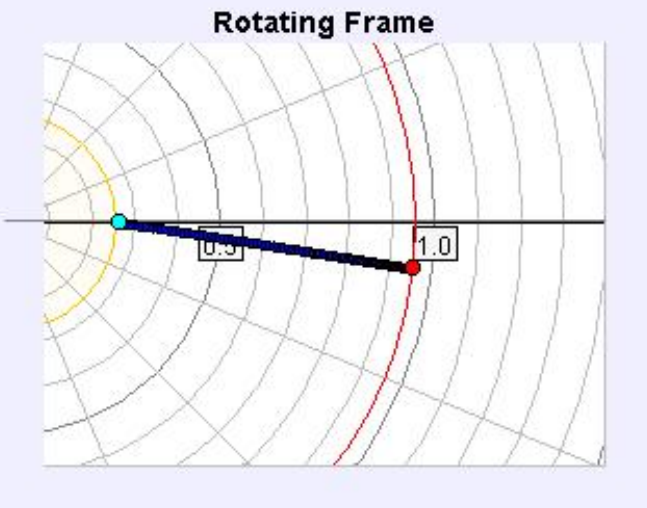


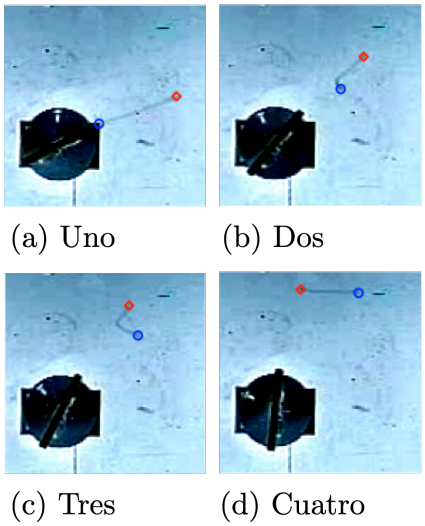
Figure 2: Coordinate system of the stretched spring attached to a rotating rotor of radius r_0. The displacement u represents the extension of a mass element from its rest position x.

Ideally, the spring extension depends on the centrifugal force, which scales with ω^2 (see Figure 2). During rotation, air drag may cause the spring to lag behind the radial line by an angle α (drag angle), and gravity may cause it to sag by an angle β (droop angle).
At $t = 0$, the inner end is released. The information that the inner end is free travels outwards as a longitudinal elastic wave. The time it takes for this signal to reach the outer end is the **collapse time**, t_c. This creates a characteristic "chevron" shape during the collapse.
A high-speed camera tracks:

- $\theta_{\text{out}}(t)$: Angular position of the outer end.
- $\theta_{\text{in}}(t)$: Angular position of the inner end.
- R_{out}: Steady-state radial distance.



(a) Drag angle α before release.



(b) Chevron shape during collapse.

Figure 3: Visualizing the spring dynamics. (a) The drag angle α due to air resistance. (b) The characteristic "chevron" shape forming after release as the wave travels outward.

4 Experimental Procedure

Part 1: Kinematics and Collapse Time (Dataset A)

The following data (Table A1) corresponds to a release event at nominal $\omega \approx 16$ rad s^{-1}.

Table A1. Angular positions from video tracking.

t (s)	θ_{out} (rad)	$\sigma_{\theta,\text{out}}$ (rad)	θ_{in} (rad)	$\sigma_{\theta,\text{in}}$ (rad)
0.00	0.143	0.10	0.005	0.12
0.05	0.835	0.10	0.487	0.12
0.10	1.608	0.10	0.641	0.12
0.15	2.251	0.10	0.533	0.12
0.20	3.178	0.10	0.703	0.12
0.25	3.972	0.10	0.388	0.12
0.30	4.793	0.10	0.276	0.12
0.35	5.550	0.10	0.282	0.12
0.40	6.396	0.10	0.377	0.12
0.45	7.034	0.10	0.193	0.12
0.50	7.447	0.10	0.022	0.12
0.55	8.103	0.10	0.019	0.12
0.60	8.583	0.10	0.127	0.12
0.65	9.129	0.10	0.030	0.12

1. Make a plot of $\theta_{\text{out}}(t)$ and $\theta_{\text{in}}(t)$ versus t. Also plot the residuals of the outer end position versus time (subtracting the early-time linear fit).

2. Using Table A1 for $t \leq 0.40$ s, perform a linear regression $\theta_{\text{out}}(t) = \theta_0 + \omega t$. Report the angular velocity ω and its uncertainty.

3. Using Table A1 for $t \geq 0.50$ s, fit a second line $\theta_{\text{out}}(t) = \theta_1 + \omega_2 t$. Explain physically why $\omega_2 < \omega$.

4. Determine the collapse time t_c as the intersection of the two fitted lines. Interpret this time in terms of signal propagation (see Figure 3b).

5. Analyze the plot of $\theta_{\text{in}}(t)$ and explain the rapid deviation from uniform rotation upon release.

Part 2: Quasi-static Equilibrium (Dataset B)

Table B1 presents steady-state measurements for different angular speeds.

Table B1. Steady-state quantities and collapse time.

ω (rad/s)	R_{out} (m)	σ_R (m)	α (deg)	σ_α (deg)	β (deg)	σ_β (deg)	t_c (s)	σ_{t_c} (s)
8	0.1088	0.002	4.34	0.5	19.52	0.7	0.4287	0.01
10	0.1399	0.002	6.58	0.5	18.14	0.7	0.4321	0.01
12	0.1742	0.002	9.28	0.5	17.79	0.7	0.4221	0.01
14	0.2241	0.002	12.98	0.5	17.81	0.7	0.4273	0.01
16	0.2699	0.002	17.28	0.5	15.95	0.7	0.4281	0.01
18	0.3291	0.002	20.46	0.5	15.71	0.7	0.4249	0.01
20	0.3958	0.002	24.55	0.5	14.47	0.7	0.4205	0.01

6. Plot R_{out} versus ω^2. Perform a linear regression $R_{\text{out}} = R_* + S\,\omega^2$. Report the slope S and intercept R_* with uncertainties.

7. Explain what determines R_* (intercept) and S (slope). Explain why the dependence is on ω^2 (refer to Figure 2).

8. Show that $S \propto R_*\, M/K$ (up to a geometric factor of order unity) and check that both sides have dimensions of $\mathrm{m\,s^2}$. State the approximations used.

Part 3: Drag and Droop Analysis

9. Using Table B1, plot α vs ω and α vs ω^2. Determine which model fits better ($\alpha \propto \omega$ or $\alpha \propto \omega^2$). Explain physically why quadratic drag is expected.

10. Plot β vs ω. Explain why β decreases as ω increases.

Part 4: Wave Propagation Analysis

11. Test if t_c in Table B1 is constant (independent of ω). Calculate the weighted mean $\bar{t}_c$.

12. Using dimensional analysis, argue that $t_c \sim \sqrt{M/K}$.

Part 5: Advanced Analysis

13. Propose a refined drag model (e.g., linear-quadratic crossover) and predict its effect on the α vs ω plot.

14. Discuss how the tension profile $T(x)$ changes with rotation and if this could introduce a weak ω-dependence in t_c.